1 Summary

Fuselage Section 13 — Forward — Door Surround Structure

Standardisation of the Fwd Cargo Door Surround Corner Fittings in Titanium Alloy

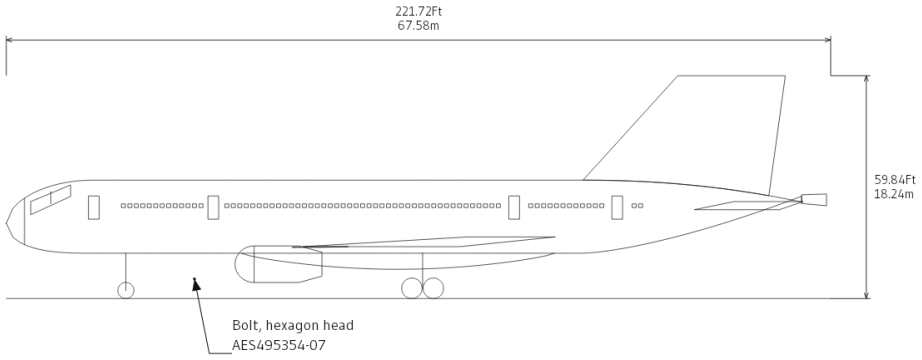
Object code	DMC-MRD-A-53-21-13-00A-040A-A	**Issue**	002-00 (revised)
Issue date	2021-06-30	**Language**	en-GB
Responsible partner	AERALIS S.A.S. (U8025)	**Originator**	AERALIS S.A.S. — Design Office, Neuhafen (U8025)
Security	01 — Unclassified	**Quality assurance**	First verification (tabtop)
Applicability	MERIDIAN-42 and MERIDIAN-46, all aircraft		

1 Summary

The four aluminium-lithium corner fittings of the forward cargo door cut-out are replaced by machined titanium items with a thicker flange and a larger load introduction radius, raising the fatigue allowable under the high-density freight spectrum.

2 Reason

Re-substantiation of the forward cargo door surround against the high-density freight loading spectrum introduced for combi and heavy-cargo operators showed a reserve factor shortfall at the lower corner fittings. Crack initiation was predicted between 34 000 and 42 000 flight cycles against a design service goal well above that. No in-service finding existed at launch; the change was driven entirely by the revised spectrum and by the fleet leader projection presented at the 2017 structures review.



Fuselage Section 13 — Forward — Door Surround Structure — location, side view — Figure 1

3 Description

The four aluminium-lithium fittings at the lower forward and lower aft corners of the cargo door cut-out are replaced by machined Ti-6Al-4V items of the same external envelope but with a 2.4 mm thicker attachment flange and a 9 mm fillet radius at the load introduction point. The web assembly D5321003375416 is retained and is re-drilled locally to the new pattern; the joint is closed with titanium bolts AES495354-07 and countersunk washers AES963502K7 in place of

4 Effects

the previous protruding-head set. Revised detail is carried on drawings D53211308437 and D53211167819. Access is through the forward cargo compartment in zones 113 and 115, with the lower shell lining removed locally.

Neither the door itself nor the door actuation installation is touched.

4 Effects

Mass increase of 3.1 kg per aircraft, symmetric about the centre line, with a centre-of-gravity shift below the reporting threshold. No electrical load change. The fittings are not interchangeable with the previous standard because the fastener pattern differs, and a mixed installation on one door surround is not permitted. Embodiment removes the 3 000 flight cycle repeat detailed visual inspection introduced by the original substantiation, which is the main operating benefit.

Illustrated parts content for ATA 53-21 and the associated removal task require reissue. The substantiation method behind this change — a load-path redistribution argument taken at the lower corner rather than a gross-area allowable — has since been read across to the freighter main deck cargo door case of MOD-160560. Nothing in the parts list, the fastener standard or the drawing set is common between the two, and the read-across is of the method only.

5 Embodiment note

Cut in on the production line at MSN M00300 in August 2020. Delivered aircraft are brought to the standard through Service Bulletin MRD-53-1012, which is preceded by the one-off inspection of Service Bulletin MRD-53-1008. Kits are held as spares under the same reference and typical embodiment is four days inside a scheduled C-check. The design justification now carries the read-across decision to MOD-160560 as a recorded item, so that the freighter substantiation is not re-argued from first principles at the next revision of either dossier.

6 Modification data

Table 1 Modification data

Modification number	MOD-160142
Class	Major
Change category	Structure
Trigger	Certification
Customer driven	No
Certification impact	Major change
Authority	EASA, FAA
Certification basis	CS 25.571, CS 25.365
Flight test required	No
Mass change	3.1 kg
Balance	Symmetric installation, CG shift below 0.02 % MAC
Electrical load change	0 VA
Interchangeability	Not interchangeable
Estimated man-hours	104

7 Applicability and effectivity

Estimated cost	455 kEUR
Lead time	30 weeks
Risk level	Low

Table 2 Parts affected

Part number	Action
AES495354-07	Added
AES963502K7	Added
D5321003375416	Modified

Cut-in at MSN M00300, absorbed inside the existing section 13 join sequence

7 Applicability and effectivity

MERIDIAN-42 and MERIDIAN-46, all aircraft

Table 3 Embodiment

Status	In service
Embodiment points	Production, Retrofit, Spares
Production effectivity	M00300
Production cut-in	2020-08-19
Retrofit through	MRD-53-1008, MRD-53-1012

8 Referenced documents

Table 4 Referenced documents

Document	Type	Relationship
MRD-53-1008	SB	Embodied by
MRD-53-1012	SB	Embodied by
MOD-160560	MOD	References
IPC-M-53-21-13-FIG-01	IPC	Impacts
AMM-M-53-21-13-000-001-A	AMM	Impacts